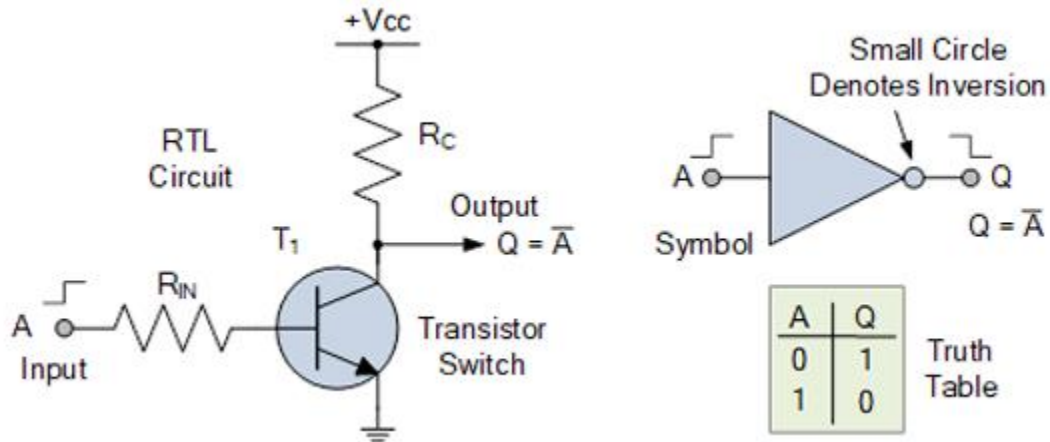


BJT Logic Gates

ENGS152 Circuits Lab

10 April 2026

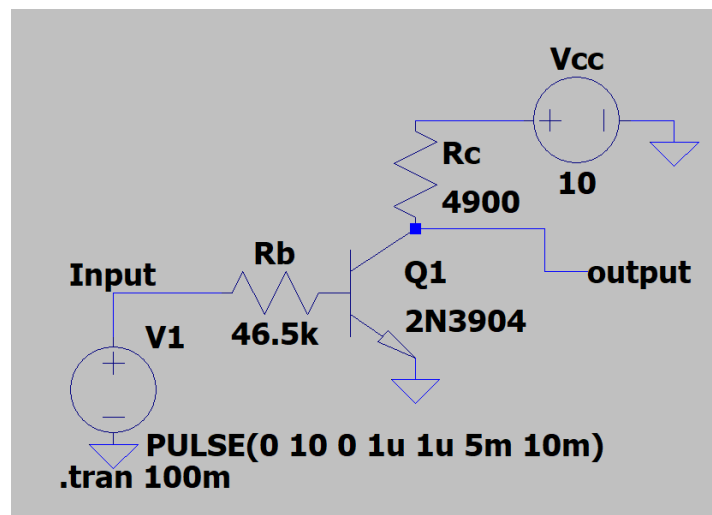
Step 1: Inverter (NOT) Gate

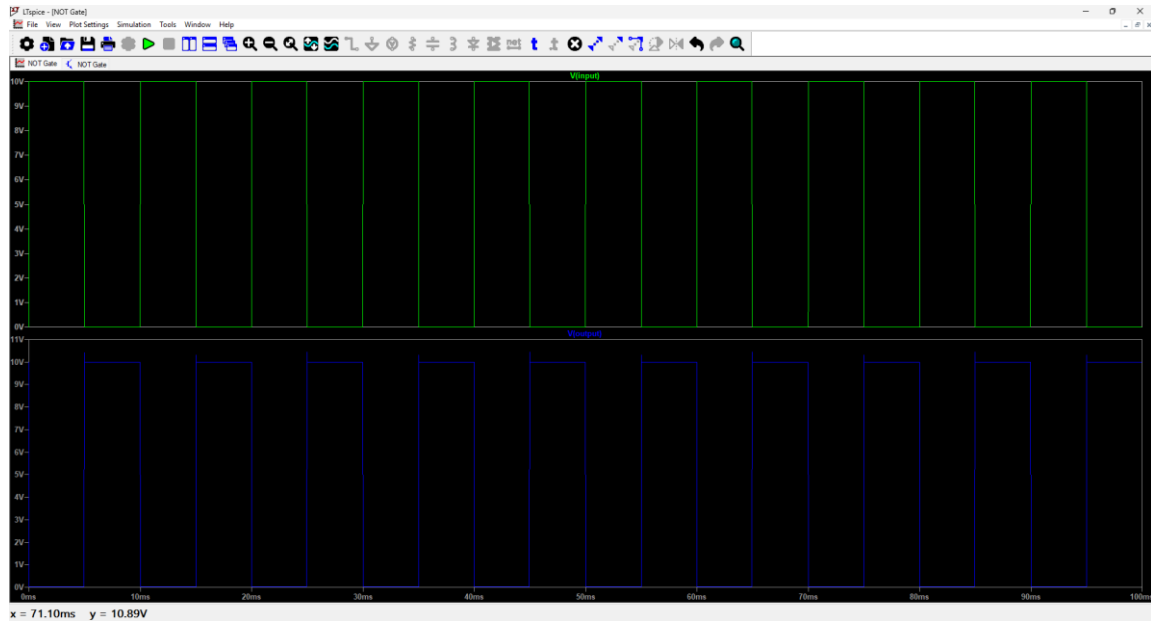


$$R_c = \frac{V_{cc} - V_c}{I_c} = \frac{9.8 \text{ V}}{2 \text{ mA}} = 4900 \, \Omega$$

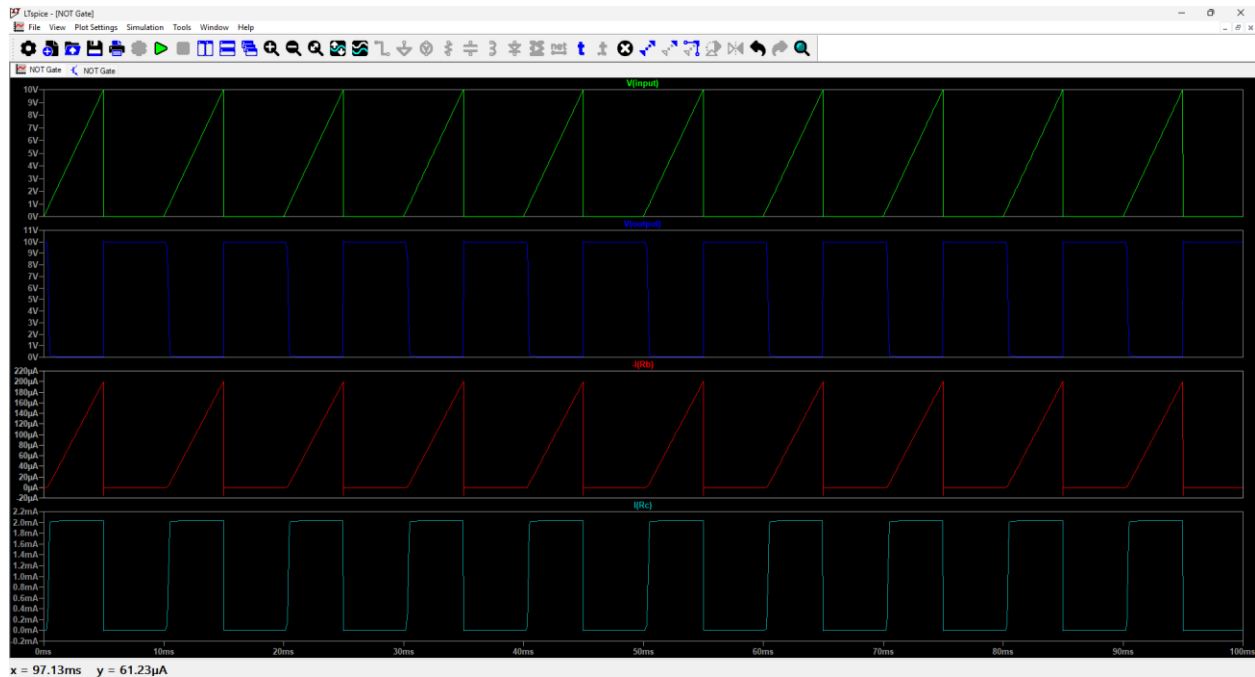
$$I_B = \frac{I_c}{\beta} = \frac{2 \text{ mA}}{10} = 0.2 \text{ mA}$$

$$R_B = \frac{V_{in} - V_{BE}}{I_B} = \frac{9.3 \text{ V}}{0.2 \text{ mA}} = 46.5 \text{ k}\Omega$$

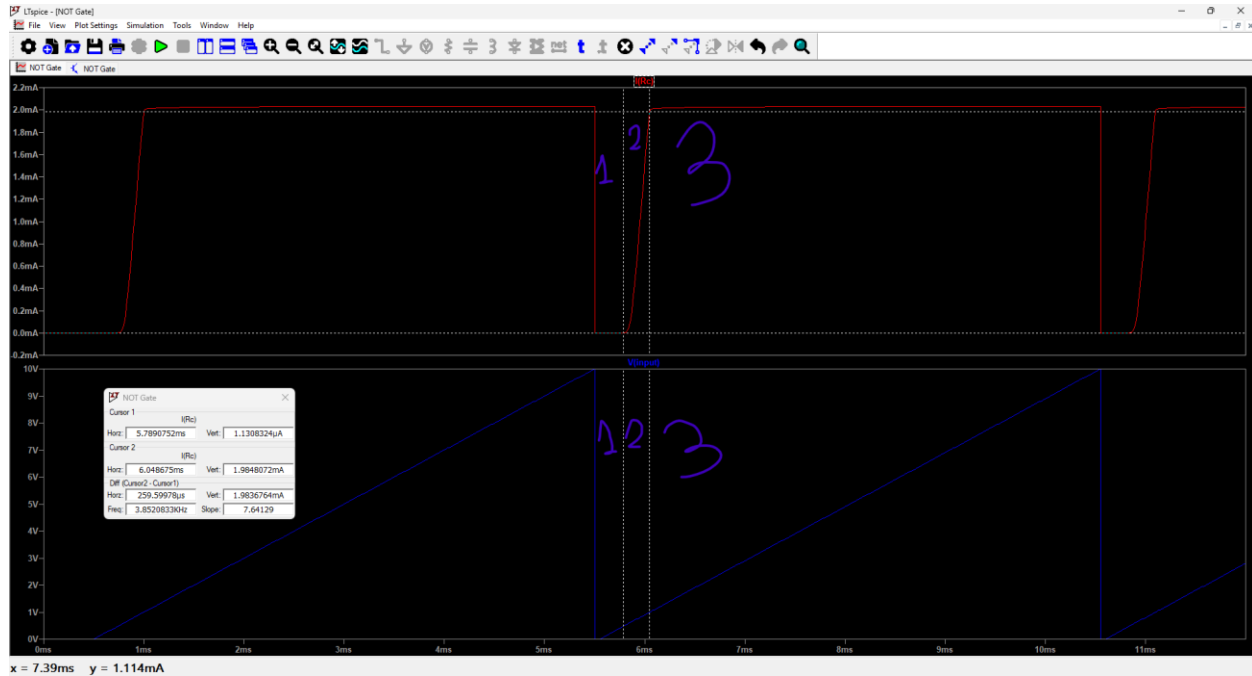




As we can see, the signal gets inverted as it passes through the NOT gate. Next up, we are asked to simulate a triangle wave and find the regions on the plot where the BJT is in the Cutoff, Linear, and Saturation modes.

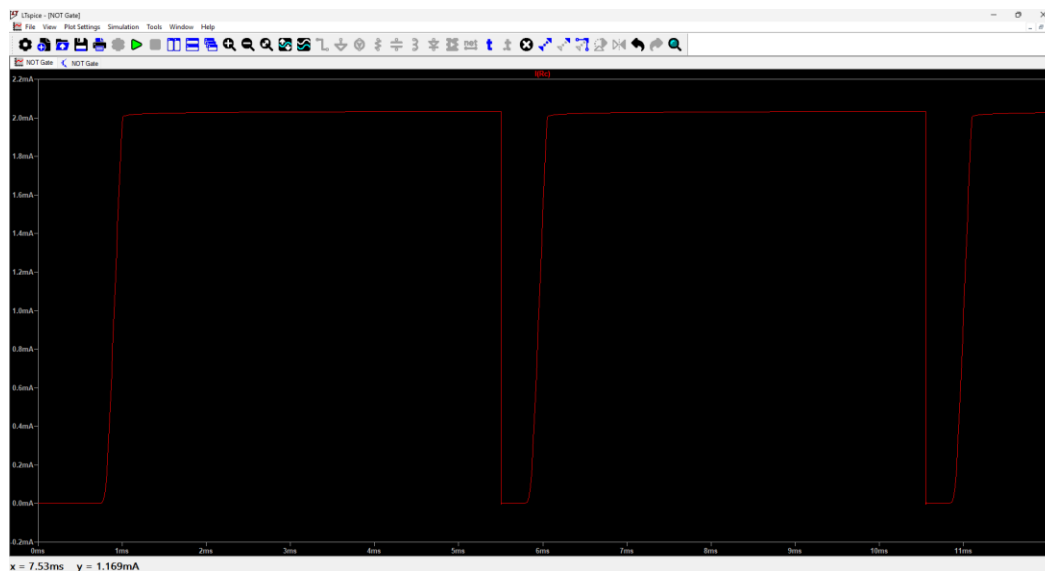


You can find the corresponding Cutoff (1), Active (linear) (2), and Saturation (3) mode regions on the I_c plot below:

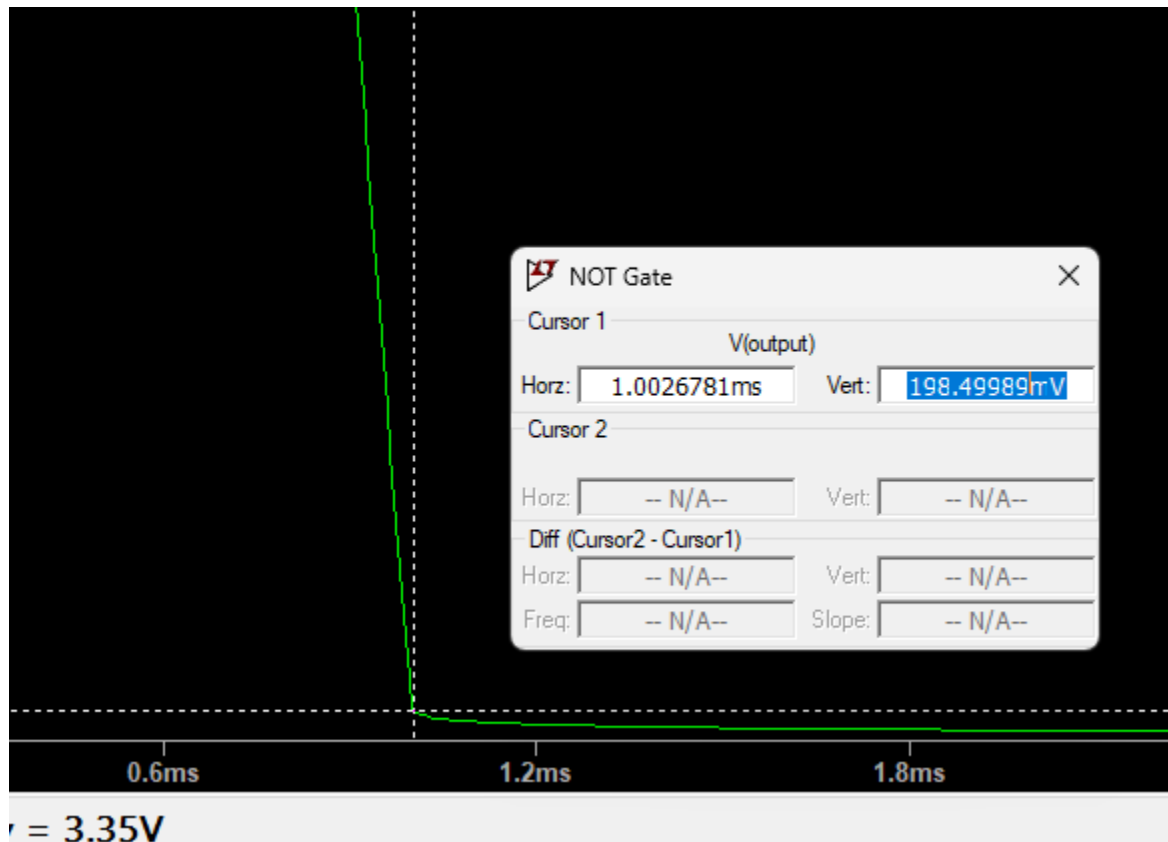


As we can see initially the current in the collector is 0. As the voltage increases and reaches 0.7 V, the BJT from cutoff mode enters the linear mode where the current in the collector and base is linear with the slope of h_{FE} , then as voltage increases till 0.8 V the transistor fully closes the circuit and acts as a switch on HIGH setting representing the saturation mode.

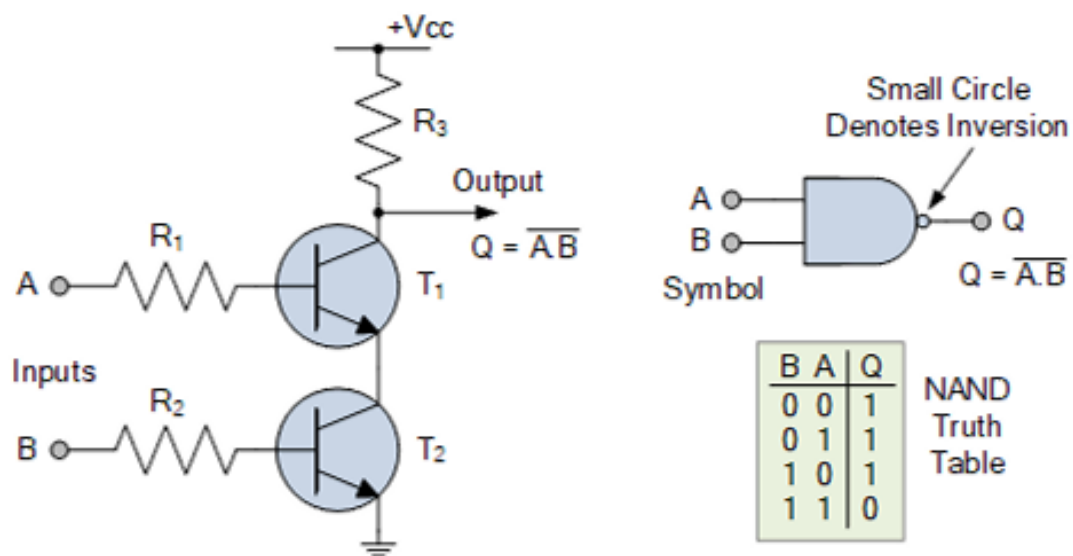
Below you can see the I_C plot, the current is 2 mA, which proves the DR3 statement.



DR5 is held because when the transistor is turned ON, the collector-emitter voltage drops until 0.2V, making the BJT enter saturation mode.



Step 2: NAND gate



$$R_c = \frac{V_{cc} - V_c}{I_c} = \frac{9.8 \text{ V}}{2 \text{ mA}} = 4900 \Omega$$

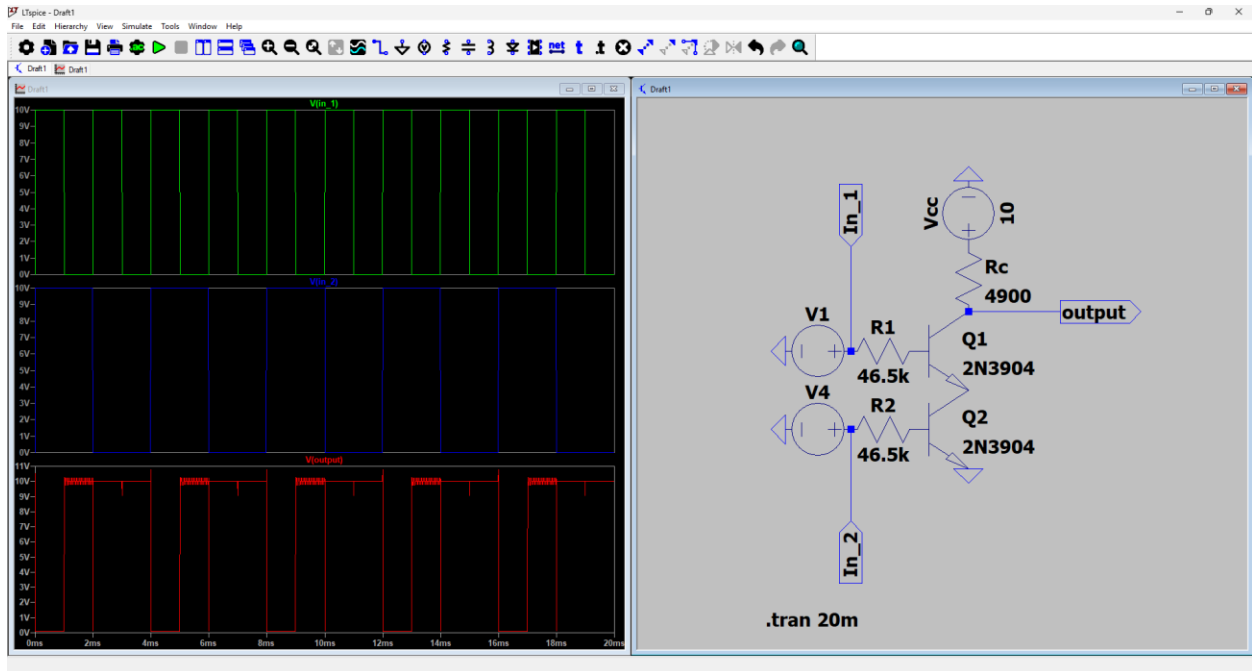
$$I_{C1} = I_{C2} = 2 \text{ mA}$$

$$I_{B1} = I_{B2} = \frac{I_c}{\beta} = \frac{2 \text{ mA}}{10} = 0.2 \text{ mA}$$

$$R_{B1} = R_{B2} = \frac{V_{in} - V_{BE}}{I_B} = \frac{9.3 \text{ V}}{0.2 \text{ mA}} = 46.5 \text{ k}\Omega$$

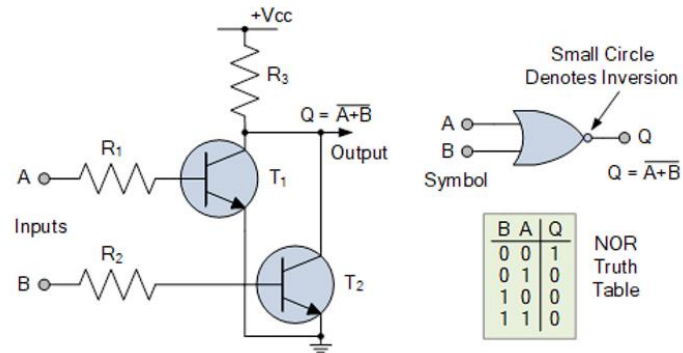
There are no differences in calculations compared to the inverter, because each transistors input is designed independently and they have to satisfy the same design requirements.

Below we can see that the NAND gate is working as expected:



B	A	Q
0	0	1
0	1	1
1	0	1
1	1	0

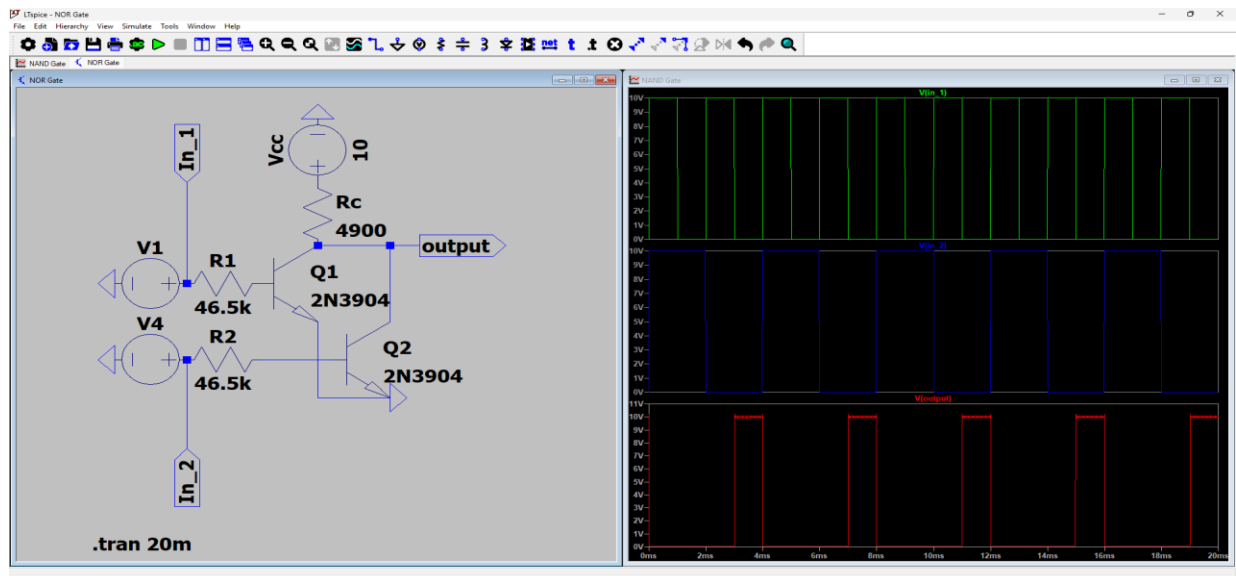
Step 3: NOR gate



$$R_c = \frac{V_{cc} - V_c}{I_c} = \frac{9.8 \text{ V}}{2 \text{ mA}} = 4900 \, \Omega, \quad I_c = 2 \text{ mA}, \quad I_{B1} = I_{B2} = \frac{I_c}{\beta} = \frac{2 \text{ mA}}{10} = 0.2 \text{ mA}$$

$$R_{B1} = R_{B2} = \frac{V_{in} - V_{BE}}{I_B} = \frac{9.3 \text{ V}}{0.2 \text{ mA}} = 46.5 \text{ k}\Omega$$

Finally, the NOR gate:



A	B	Q
0	0	1
0	1	0
1	0	0
1	1	0

